

Umami is one of five basic tastes that humans recognize, and it is triggered by L-amino acids as well as some synthetic ingredients and organic acids. Three common umami tastant molecules are shown in Figure 1.

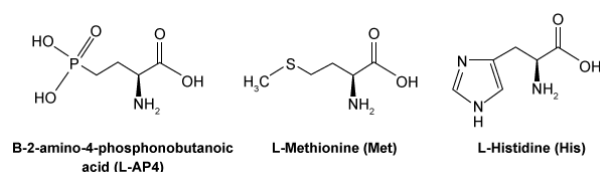


Figure 1 Three common tastant molecules known to trigger the umami taste sensation

Umami taste signals are triggered upon binding of tastant molecules to the G protein–coupled receptor (GPCR) heterodimer T1R1-T1R3, which ultimately causes the breakdown of phosphatidylinositol 4,5-bisphosphate (PIP₂) into the messenger molecules inositol-1,4,5-trisphosphate (IP₃) and diacylglycerol (DAG). IP₃ triggers the release of Ca²⁺ cations and the subsequent gating of the taste-transduction channels (Figure 2).

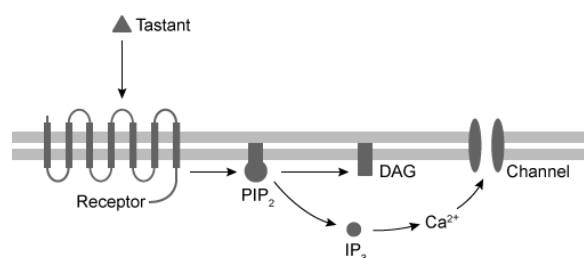


Figure 2 Signal pathway for transduction of umami taste sensation

Inosine monophosphate (IMP) has been suggested as enhancing (potentiating) the response of T1R1+T1R3 GPCRs to umami tastant molecules. Scientists tested this hypothesis by culturing receptors from human embryonic kidney (HEK) cells containing promiscuous G proteins. The HEK cells were then stimulated with umami tastants in the presence or absence of IMP. In each test, the cells were exposed to the same amounts of amino acid tastant and IMP. The number of cells that responded to each tastant was measured using a calcium indicator dye (FURA-2AM) in conjunction with fluorescence microscopy. By comparing the ratio of the HEK cells that were stimulated in the presence of IMP to those that were stimulated in the absence of IMP, the relative extent of potentiation can be assessed. The results are shown in Figure 3.

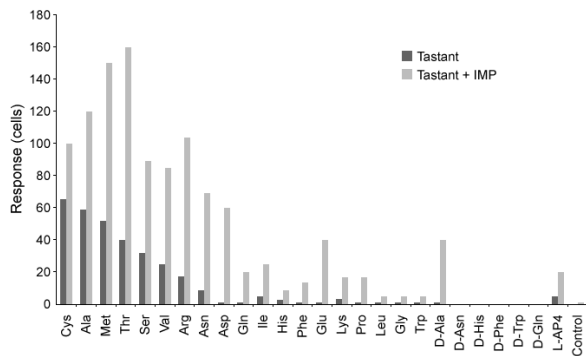


Figure 3 Quantification of amino-acid responses for T1R1+T1R3 with and without IMP

Which of the following statements is true regarding the bonds in the tastant molecule L-methionine?

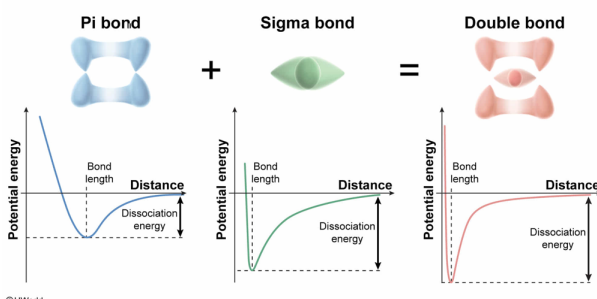
- I. The π bond component in a C=O bond requires less energy to break than its σ bond component.
- II. The C=O double bond is stronger than the C–O single bond in the fully protonated form (excluding effects from resonance).
- III. The π bond component in a C=O bond is more stable (less reactive) than the σ bond component.

- ☐ A. I only [4%]
- ✓ ☒ B. I and II only [52%]
- ☐ C. II and III only [33%]
- ☐ D. I, II, and III [9%]

Correct

53% answered correctly

Explanation:



Covalent bonds are formed when two atoms share electrons by overlapping their

atomic orbitals. **Sigma (σ) bonds**, which are generally the first bond formed between atoms, are created when an electron pair is shared between the two atoms by an end-to-end overlap of their atomic orbitals along the internuclear axis. Because of the **efficient overlap** of the atomic orbitals, σ bonds tend to be **stronger** and very **stable** (less reactive).

In contrast, **pi (π) bonds** are created by the side-to-side overlap of p orbitals along a plane perpendicular to (ie, above and below) the internuclear axis. Because the overlap in electron density is not as efficient as for σ bonds, π bonds are **weaker** and **less stable** (more reactive) than σ bonds. As a result, π bonds require less energy to break (ie, they have a **smaller dissociation energy**) than σ bonds (**Number I**).

After a σ bond forms between two atoms, every additional bond formed between the atoms is a π bond. As such, a **double bond** between two atoms (eg, $C=C$, $C=O$) consists of both a σ bond and a π bond component. Consequently, the dissociation energy of a double bond is the combined energy required to break both the σ bond and the π bond components. As such, a **double bond** between two atoms is generally **stronger than a single bond** between the same two atoms (**Number II**) when any effects of resonance are excluded.

Therefore, only statements I and II are true regarding the bonds in the tastant molecule L-methionine.

(Number III) This statement is incorrect because the π bond component in $C=O$ is weaker than the σ bond component, causing the π bond component to be *less stable* (*more reactive*).

Educational objective:

Sigma (σ) bonds are stronger and more stable than pi (π) bonds, which gives σ bonds a higher dissociation energy than π bonds. Sigma bonds form first between two atoms, and every bond formed thereafter is a π bond. Although individual π bonds are weaker than σ bonds, a double bond is composed of both a σ bond and a π bond, and therefore a double bond is stronger than a single bond between the same atoms.

Time spent:0 sec

Subject:

General Chemistry

QID:400434

System:

Interactions of Chemical Substances

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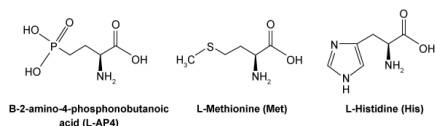


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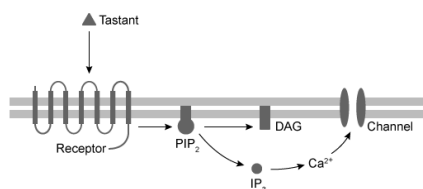


Figure 2 Signal pathway for transduction of umami taste sensation

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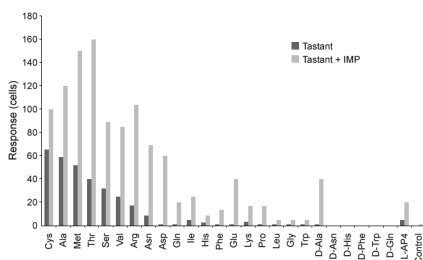
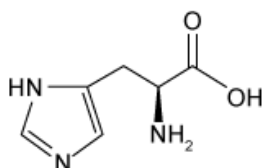


Figure 3 Quantification of amino-acid responses for T1R1+T1R3 with and without IMP

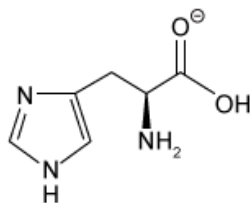
Which of the following is an acceptable resonance structure for the umami tastant molecule L-histidine shown in Figure 1?

☐ A.



[31%]

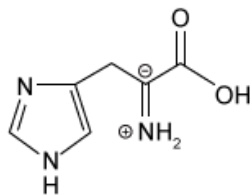
☐ B.



50%

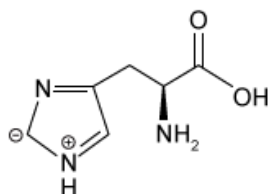
[3%]

○ C.



[9%]

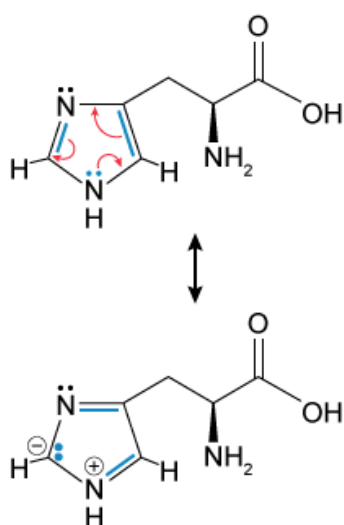
✔ ● D.



[54%]

Correct

54% answered correctly

Explanation:**Resonance structure of L-histidine**

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Molecules can be represented by more than one valid **Lewis structure** if their electrons can be **delocalized** (ie, the electrons can occupy different locations among the atoms in the structure via **resonance**). The collection of these Lewis structures is known as **resonance structures**.

The correct answer adheres to the following rules:

- Atoms never move, **only electrons**.
- All resonance structures must have the same total **number of valence electrons**.

- The **octet rule** must be obeyed for **first- and second-row elements**.
- Only electrons in pi (π) bonds (eg, from double or triple bonds) or nonbonding lone pairs can move, **not electrons in sigma (σ) bonds** (ie, single bonds).
- In general, **electron movement** should only be **to adjacent atoms** when going from one resonance structure to another.
- The **overall charge** of the molecule **must not change**; however, the formal charges of the constituent atoms can change.

Among the given structures, only Choice D follows the rules above.

(Choice A) The hydrogen atom on the imidazole ring has moved, making a **constitutional isomer** rather than a resonance structure.

(Choice B) An electron has been added to the molecule. Per the rules above, the number of valence electrons should be the same for all resonance structures. Furthermore, the addition of an electron has changed the overall charge of the molecule from neutral to negative, which is not allowed.

(Choice C) The octet rule has been broken because the carbon atom now has 10 electrons. Per the rules above, the octet rule must always be obeyed for first- and second-row elements.

Educational objective:

Resonance structures are Lewis structures of a given molecule in which electrons have been delocalized. Acceptable resonance structures move only electrons and generally only to adjacent atoms, keep the number of valence electrons constant, obey the octet rule, do not break sigma (σ) bonds, and keep the overall charge of the molecule constant.

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Subject:
 General Chemistry

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System:
 Interactions of Chemical Substances

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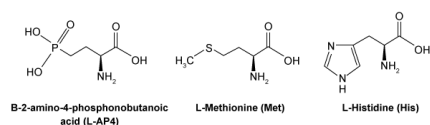


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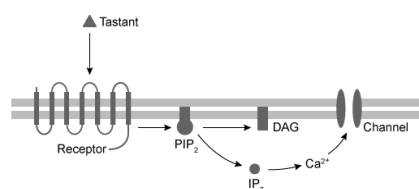


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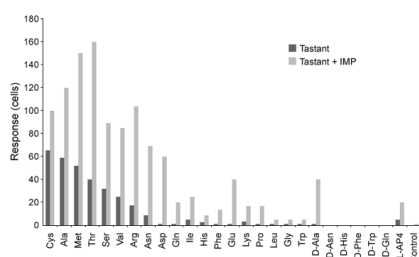
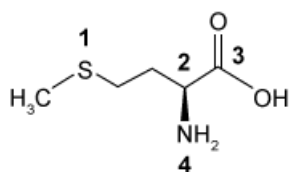
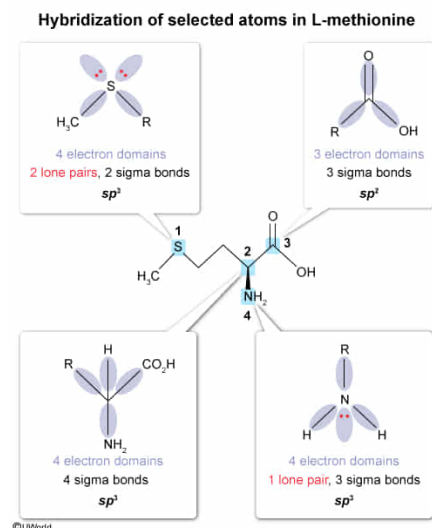


Figure 3 Quantification of amino-acid responses for T1R1+T1R3 with and without IMP

Which of the following correctly lists the respective hybridization of each of the numbered atoms (in numerical order) for the tastant molecule L-methionine shown below?



- ☐ A. sp , sp^2 , sp^2 , sp^2 [10%]
☐ B. sp , sp^3 , sp^2 , sp^2 [19%]
☐ C. sp^3 , sp^3 , sp^3 , sp^3 [2%]
☒ D. sp^3 , sp^3 , sp^2 , sp^3 [67%]

Explanation:

Hybrid orbitals are formed by **combining** the **atomic orbitals** for a given atom to generate new **electron domains** (ie, electron-dense areas). To identify the type of hybrid orbital in an atom, determine its number of electron domains by summing the **sigma (σ) bonds** and **lone pairs** (ie, nonbonding electrons). The **sum of sand psuperscripts** on the hybrid name should match this number: sp^3 -hybridized atoms have four electron domains ($s^1 + p^3 = 4$), sp^2 have three ($s^1 + p^2 = 3$), and sp have two ($s^1 + p^1 = 2$). (The superscript 1 is implied when none is written.)

- Atom 1 (sulfur) has four electron domains (ie, two σ bonds and two lone pairs). Therefore, its hybridization is sp^3 .
- Atom 2 (carbon) has four σ bonds, giving it a hybridization of sp^3 .
- Atom 3 (carbon) has three electron domains (ie, two single bonds and one double bond). The double bond counts as a single electron domain (ie, the pi (π) bond is not counted in determining hybridization). Consequently, atom 3 is sp^2 hybridized.
- Atom 4 (nitrogen) has three σ bonds and one lone pair for a total of four electron domains and a hybridization of sp^3 .

Therefore, the correct list of the respective hybridization of each of the numbered atoms (in numerical order) for the tastant molecule L-methionine is sp^3 , sp^3 , sp^2 , sp^3 (Choice D).

(Choice A) Structural drawings of organic molecules do not always explicitly show every C–H bond or lone pair of electrons in the molecule. Atoms 1, 2, and 4 are given incorrect hybridizations by not accounting for the lone pairs around S and N and by not recognizing the C–H bond of atom 2.

(Choice B) The 4 electron domains of atom 1 (ie, 2 lone pairs and 2 σ bonds) and the 4 electron domains of atom 4 (ie, 1 lone pair and 3 σ bonds) give both of these atoms an sp^3 hybridization.

(Choice C) Atom 3 was given an extra electron domain by counting the π bond on its double bond (ie, counting the C=O bond as two electron domains instead of counting it as one domain).

Educational objective:

Hybrid orbitals are formed by combining the atomic orbitals of an atom. The hybridization of an atom is determined by counting the electron domains (ie, sigma (σ) bonds and lone pairs) and assigning a hybrid name. The sum of superscripts must equal the number of electron domains. Structural drawings of organic molecules do not always explicitly show every C–H bond or lone pair of electrons in the molecule.

Time spent: 0 sec
Subject:
General Chemistry

QID: 400436
System:
Interactions of Chemical Substances

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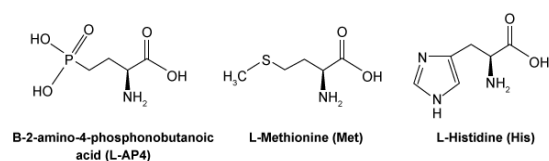


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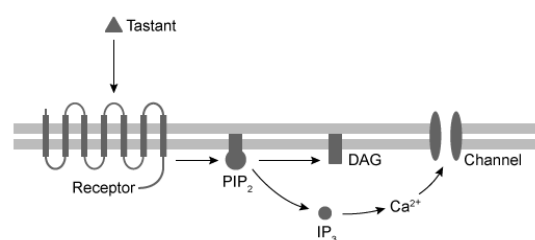


Figure 2 Signal pathway for transduction of umami taste sensation

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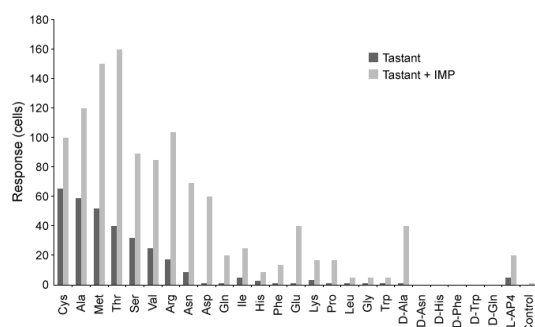
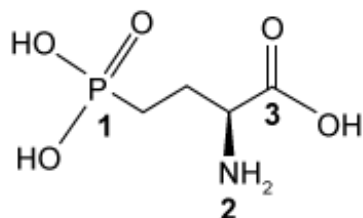


Figure 3 Quantification of amino-acid responses for T1R1+T1R3 with and without IMP

What are the molecular geometries of the groups bonded to the central atoms 1, 2, and 3, respectively, in the tautomer molecule L-AP4 shown below?

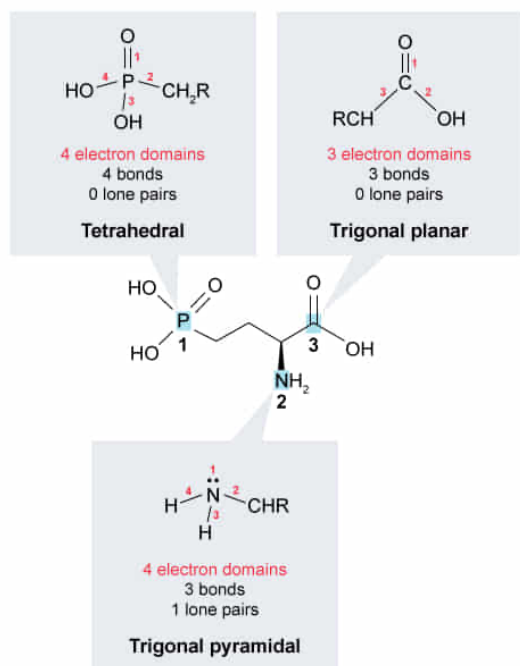


- ✓ ☒ A. Tetrahedral, trigonal pyramidal, trigonal planar [71%]
☐ B. Tetrahedral, trigonal planar, trigonal planar [18%]
☐ C. Trigonal pyramidal, trigonal planar, bent [5%]
☐ D. Trigonal pyramidal, trigonal pyramidal, trigonal pyramidal [4%]

Correct

71% answered correctly

Explanation:



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Valence shell electron pair repulsion (VSEPR) theory states that a molecule's shape is determined by distributing electrons as far apart as possible to **minimize electron repulsion** and maximize attractive forces.

To identify the geometry of a molecule, first determine the molecule's **electron group geometry** by counting the number of **electron domains**, which includes **lone electron pairs** and **sigma (σ) bonds**. Next, determine how many of the electron domains are **bonds** and how many are

lone pairs. If all the electron domains are from bonds, the **molecule's geometry** will be the same as its **electron group geometry**. If **lone pairs** are present, the molecule's geometry will be different from electron group geometry.

In the molecule L-AP4, the highlighted phosphorus and carbon atoms have only bond domains and will be **tetrahedral** and **trigonal planar**, respectively. Nitrogen has four electron domains with one lone pair and will be **trigonal pyramidal**.

(Choice B) For atom 2 (nitrogen) to be trigonal planar, it would need three electron domains distributed as three bonds.

(Choice C) For atom 1 (phosphorous) to be trigonal pyramidal, it would need four electron domains distributed as three bonds and one lone pair. For atom 3 (carbon) to be **bent**, it would need three electron domains in the form of two bonds and one lone pair.

(Choice D) For atom 3 (carbon) to be trigonal pyramidal, it would need four electron domains distributed as three bonds and one lone pair.

Educational objective:

Valence shell electron pair repulsion (VSEPR) theory states that the distribution of bonds and lone pairs will minimize repulsion between electrons. To find the geometry of a molecule, count the number of electron domains and determine the electron group geometry. If lone pairs are present, the molecule's geometry (determined by the orientations of bonded atoms) will be different from the electron group geometry.

Time spent: 0 sec

Subject:
General Chemistry

QID: 400437

System:
Interactions of Chemical Substances

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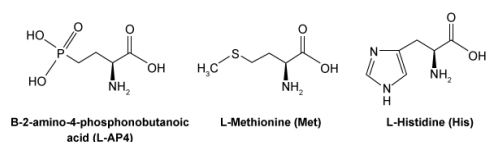


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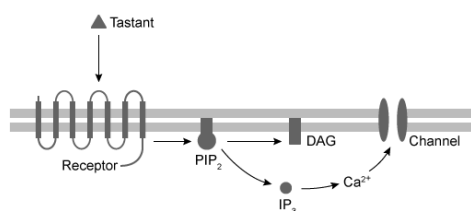


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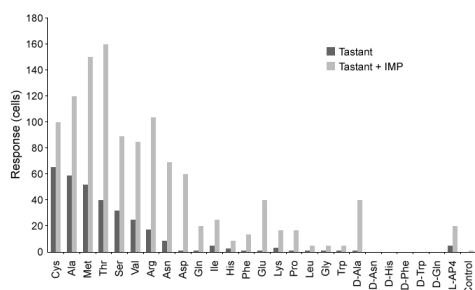


Figure 3 Quantification of amino-acid responses for T1R1+T1R3 with and without IMP

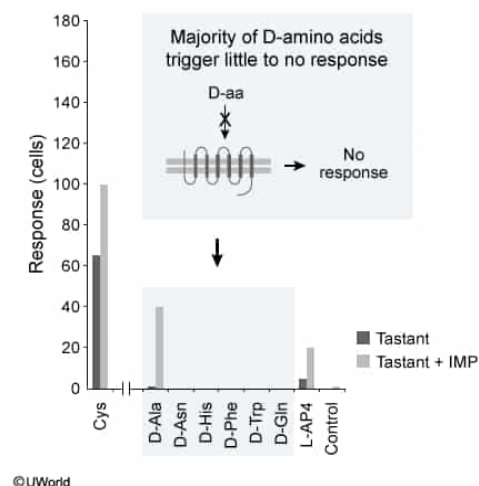
Which of the following best explains the results regarding D-amino acids in Figure 3?

- ☐ A. D-amino acids are not as abundant in HEK cells as L-amino acids. [17%]
- ☐ B. D-amino acids do not fluoresce upon addition of FURA-2AM dye. [4%]
- ☐ C. D-amino acids trigger the formation of DAG. [1%]
- ☒ D. Most D-amino acids do not effectively bind to the GPCR heterodimer T1R1+T1R3. [76%]

Correct

76% answered correctly

Explanation:



The umami sensation is triggered by the **binding of tastant molecules** to the GPCR T1R1+T1R3 heterodimer, ultimately leading to the release of Ca^{2+} ions. The number of cells that respond to a given tastant was determined using a calcium dye binding assay with fluorescence microscopy. As shown in Figure 3, the y-axis indicates **cell response**, and it is evident that **D-amino acids** elicit only a small response, if any; of the D-amino acids, only D-Ala caused a noticeable cell response. Because responses are caused by tastant binding, it can be concluded that a majority of D-amino acids do not bind effectively to T1R1+T1R3 GPCR cells.

(Choice A) According to the experiment, each cell was exposed to an equal amount of both L-amino acids and D-amino acids.

(Choice B) The FURA-2AM dye causes fluorescence when in the presence of calcium, not amino acids.

(Choice C) According to the passage, IP_3 and DAG are produced together from the same reaction (ie, the breakdown of PIP_2). If D-amino acids triggered DAG formation, the production of IP_3 and the subsequent calcium release and cellular response would be detected.

Educational objective:

Interpretation of graphic data requires an understanding of the underlying mechanisms. Cells may have altered responses to stimuli for a number of reasons. In this case, amino acids induce a response by binding to a receptor, so failure to induce a response indicates a lack of binding.

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General Chemistry

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System:
Interactions of Chemical Substances

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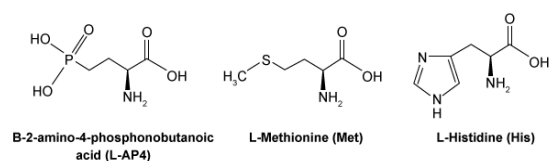


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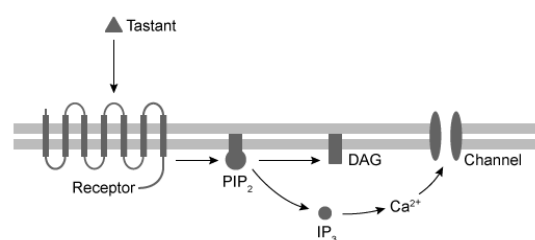


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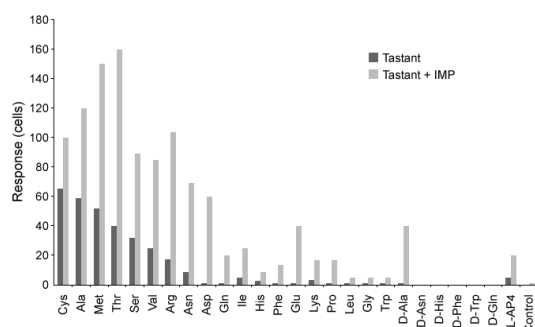


Figure 3 Quantification of amino-acid responses for T1R1+T1R3 with and without IMP

Which of the following statements is true concerning the results shown in Figure 3?

- ☐ A. For the compounds potentiated by IMP, the extent of potentiation is five-fold. [3%]
- ☐ B. For the compounds potentiated by IMP, the extent of potentiation is equal.
- ☐ C. IMP most effectively potentiates Thr. [43%]
- ✓ ☒ D. IMP most effectively potentiates Asp. [52%]

Correct

52% answered correctly

Explanation:

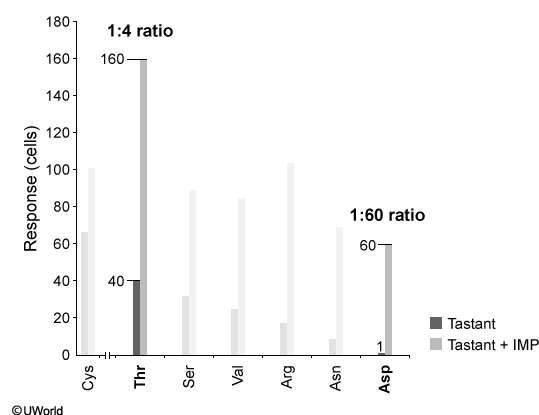


Figure 3 shows that some molecules exhibit a greater overall response to tastants than others, as well as a greater difference in response between tastant and tastant + IMP. The increase in the **ratio** of the response with IMP to the response without IMP is greatest for Asp. The **degree of potentiation** by IMP for a tastant molecule is defined by the relative increase in response upon addition of IMP. Therefore, IMP most effectively potentiates Asp.

(Choices A and B) Figure 3 shows that the ratio of the response of tastant + IMP to tastant alone for each molecule is different. Therefore, IMP potentiates different tastant molecules to different degrees.

(Choice C) Although the difference in responding cells for Thr with and without IMP (~120 cell difference) is much greater than for Asp (~60 cell difference), the *ratio* of the response for tastant + IMP relative to the initial condition is less for Thr (~4:1) than for Asp (~60:1).

Educational objective:

Drawing conclusions from experimental results requires careful analysis of the experimental data. The degree of potentiation, or a change in a system's response to a stimulus, is best measured as a ratio. This ratio shows the percent change in a signal's intensity in the presence of a potentiator.

Time spent: 0 sec

Subject:

General Chemistry

QID: 400439

System:

Interactions of Chemical Substances